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2.2.1 Introduction

Under the California Building Code (CBC), handrails and guards (railings) must resist a uniform load of 50 plf and a concentrated point load of 200 lbs, both applied horizontally to the top rail. Intermediate rails, balusters, and infill panels must separately withstand a concentrated load of 50 lbs.

```
  =======  <-- Top Rail
     |     <-- Lateral Load (P)
     |
     |
     |
     |  Height (h)
     |
     |
     |
     |
===========  <-- Fixed Support / Deck Surface


Structural Schematic of Railing and Loads
```

2.2.2 Railing Design

Table 1: Import Functions (rvsrc/scripts/sectprop2.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL
bending_stress_udl(ln, w, b, d)	Maximum bending stress in a simply supported beam with UDL.
nds_beam_check(** kwargs)	Check stress and deflection for a simply supported wood beam using NDS.
nds_post_check(** kwargs)	Check stress at cantilever post

Design properties.

```
Function Arguments Dictionary : post1 (units: inch, pounds)
===========================================================================
h_1 = 42.  # post height
P_1 = 200  # concentrated load
b_1 = 3.5  # post width
d_1 = 3.5  # post depth
E_1 = 1.5*(10**6)  # modulus of elasticity
F_b = 1000  # allowable bending stress
C_D = 1.0   # load duration factor
C_M = 0.85  # wet service factor
C_F = 1.0   # size factor
C_t = 1.0   # temperature factor
C_i = 0.8   # incising factor
C_r = 1.0   # repetitive member factor
```

```
C_c = 1.0   # curvature factor
C_L = 1.0   # beam stability factor
C_b = 1.0   # bearing area factor
deflect_limit = 360.0 # max allowable deflection ln_1/deflect_limit
===========================================================================
```

Design Results

Eq. 1: Check Deck Beam

```
nds_post_check | units: inch, pounds

     Post Check Results:
     ===============================
     P: 200.00
     fb: 1175.51
     Fb_prime: 680.00
     E_prime: 1275000.00
     stress_ratio: Not OK
     deflection: 1.55
     deflection_ratio:  Not OK
```